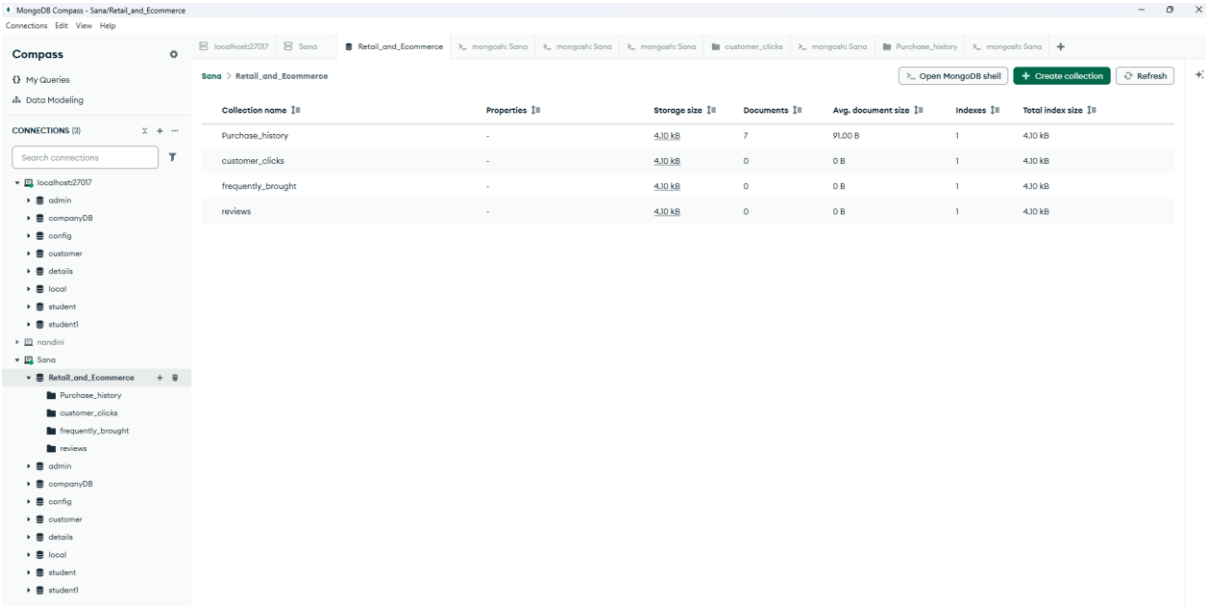
LAB 1

Domain: Retail and Ecommerce

Creating and inserting values in the table.




```
>_MONGOSH
> use Retail_and_Ecommerce
< switched to db Retail_and_Ecommerce
> db.Purchase_history.insertMany([
    { customer_id: 101, product_id: 201, quantity: 1, total_amount: 499.99 },
    { customer_id: 102, product_id: 202, quantity: 2, total_amount: 899.98 },
    { customer_id: 103, product_id: 203, quantity: 1, total_amount: 199.99 },
    { customer_id: 104, product_id: 204, quantity: 3, total_amount: 1499.97 },
    { customer_id: 105, product_id: 205, quantity: 2, total_amount: 599.98 },
    { customer_id: 106, product_id: 206, quantity: 1, total_amount: 1299.99 },
    { customer_id: 107, product_id: 207, quantity: 4, total_amount: 799.96 }
  ])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6981bf15cd41eb5d8d9fc98a'),
      '1': ObjectId('6981bf15cd41eb5d8d9fc98b'),
      '2': ObjectId('6981bf15cd41eb5d8d9fc98c'),
      '3': ObjectId('6981bf15cd41eb5d8d9fc98d'),
      '4': ObjectId('6981bf15cd41eb5d8d9fc98e'),
      '5': ObjectId('6981bf15cd41eb5d8d9fc98f'),
      '6': ObjectId('6981bf15cd41eb5d8d9fc990')
    }
  }
Retail_and_Ecommerce>
```

```
> use Retail_and_Ecommerce
< switched to db Retail_and_Ecommerce
> db.customer_clicks.insertMany([
    { customer_id: 101, product_id: 201, device: "Mobile" },
    { customer_id: 102, product_id: 202, device: "Desktop" },
    { customer_id: 103, product_id: 203, device: "Tablet" },
    { customer_id: 104, product_id: 204, device: "Mobile" },
    { customer_id: 105, product_id: 205, device: "Desktop" },
    { customer_id: 106, product_id: 206, device: "Mobile" },
    { customer_id: 107, product_id: 207, device: "Desktop" }
  ])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6981bf6e486ce2051611976d'),
      '1': ObjectId('6981bf6e486ce2051611976e'),
      '2': ObjectId('6981bf6e486ce2051611976f'),
      '3': ObjectId('6981bf6e486ce20516119770'),
      '4': ObjectId('6981bf6e486ce20516119771'),
      '5': ObjectId('6981bf6e486ce20516119772'),
      '6': ObjectId('6981bf6e486ce20516119773')
    }
  }
Retail_and_Ecommerce>
```

```
> use Retail_and_Ecommerce
< switched to db Retail_and_Ecommerce
> db.frequently_bought.insertMany([
    { product_id: 201, bought_with: 202 },
    { product_id: 202, bought_with: 203 },
    { product_id: 203, bought_with: 204 },
    { product_id: 204, bought_with: 205 },
    { product_id: 205, bought_with: 206 },
    { product_id: 206, bought_with: 207 },
    { product_id: 207, bought_with: 201 }
  ])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6981bf9bcb56cbbff72cd9fd'),
      '1': ObjectId('6981bf9bcb56cbbff72cd9fe'),
      '2': ObjectId('6981bf9bcb56cbbff72cd9ff'),
      '3': ObjectId('6981bf9bcb56cbbff72cda00'),
      '4': ObjectId('6981bf9bcb56cbbff72cda01'),
      '5': ObjectId('6981bf9bcb56cbbff72cda02'),
      '6': ObjectId('6981bf9bcb56cbbff72cda03')
    }
  }
Retail_and_Ecommerce>
```

```
> use Retail_and_Ecommerce
< switched to db Retail_and_Ecommerce
> db.reviews.insertMany([
    { customer_id: 101, product_id: 201, rating: 5 },
    { customer_id: 102, product_id: 202, rating: 4 },
    { customer_id: 103, product_id: 203, rating: 3 },
    { customer_id: 104, product_id: 204, rating: 5 },
    { customer_id: 105, product_id: 205, rating: 4 },
    { customer_id: 106, product_id: 206, rating: 2 },
    { customer_id: 107, product_id: 207, rating: 5 }
  ])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6981bfd84ffcf954f476c978'),
      '1': ObjectId('6981bfd84ffcf954f476c979'),
      '2': ObjectId('6981bfd84ffcf954f476c97a'),
      '3': ObjectId('6981bfd84ffcf954f476c97b'),
      '4': ObjectId('6981bfd84ffcf954f476c97c'),
      '5': ObjectId('6981bfd84ffcf954f476c97d'),
      '6': ObjectId('6981bfd84ffcf954f476c97e')
    }
  }
Retail_and_Ecommerce>
```

QUERIES

1.Which product generates the highest total sales in the retail and ecommerce database?

```
>_MONGOSH

> use Retail_and_Ecommerce
< switched to db Retail_and_Ecommerce
> db.Purchase_history.aggregate([
    {
      $group: {
        _id: "$product_id",
        total_sales: { $sum: "$total_amount" }
      }
    },
    { $sort: { total_sales: -1 } },
    { $limit: 2 }
  ])
< {
    _id: 204,
    total_sales: 1499.97
  }
  {
    _id: 206,
    total_sales: 1299.99
  }
```

2.Update rating of customer_id :106 for product_id: 206 from 2->3

```
> db.reviews.updateOne(
    { customer_id: 106, product_id: 206 },
    { $set: { rating: 3 } }
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
> db.reviews.find({ customer_id: 106, product_id: 206 })
< {
    _id: ObjectId('6981bfd84ffcf954f476c97d'),
    customer_id: 106,
    product_id: 206,
    rating: 3
  }
```

3. Remove review by customer_id:105 for product_id: 205

```
> db.reviews.deleteOne(
    { customer_id: 105, product_id: 205 }
  )
< {
    acknowledged: true,
    deletedCount: 1
  }
> db.reviews.find({ customer_id: 105, product_id: 205 })
<
Retail_and_Ecommerce>
```